MILITARY SPECIFICATION

CARRIER, AIR DEFENSE MISSILE SYSTEM, M975

1. SCOPE

1.1 Scope. This specification covers a full-tracked, self-propelled Carrier Vehicle for U.S. ROLAND FIRE UNIT MODULE.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids and request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

VV-F-800	-Fuel, Oil, Diesel

MILITARY

MIL-R-3065	-Rubber, Fabricated Parts
MIL-W-45205	-Welding, Inert-Gas, Metal Arc, Aluminum Alloys, Readily Weldable for Structures, Excluding Armor
MIL-W-45206	-Welding, Aluminum Alloy Armor
MIL-H-46709	-Howitzer, Light, Self-Propelled: 105MM, M108; and Howitzer, Medium, Self-Propelled: 155MM, M109, M109A1, M109A1B, M109A2; and Processing for Storage and Shipment of
MIL-E-52798	-Enamel, Alkyd, Camouflage

STANDARDS

FEDERAL

FEDERAL STD. NO. 595 - Colors.

FSC: 2350

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn: DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	-Identification Marking of U.S. Military Property
MIL-STD-193	-Painting Procedures Tactical Vehicle (Tracked and Wheeled)
MIL-STD-210	-Climatic Extremes for Military Equipment
MIL-STD-417	-Rubber Compositions, Vulcanized General Purpose, Solid (Symbols and Tests)
MIL-STD-461	-Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-642	-Identification and Marking of Administrative, Combat, and Tactical Transport Vehicles.

PUBLICATIONS

ENGINEERING PARTS LIST

LP 8750009	-Carrier, Air Defense Missile, System, M975

MASTER LIST

QM 8750009	-Master List of Supplementary Quality Assurance Provisions for Carrier, Air Defense Missile System, M975
QD 8750009	-Master List of Inspection Equipment Drawing for Carrier, Air Defense Missile System, M975

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Preproduction model. When specified (see 6.2), a preproduction model (s) of the complete vehicle to be furnished under this specification shall be produced prior to the

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manufacture or fabrication of vehicles in quantity. The vehicle (s) when complete shall be submitted to the Government for examination and tests to determine conformance to the requirements of this specification. Vehicle (s) submitted by the contractor shall be fully representative of vehicles proposed to be supplied by the vehicle manufacturer from production facilities and tooling under the supply contract.

3.2 First production vehicles. The vehicle selected shall be fully representative of vehicles proposed to be furnished under the contract, shall be submitted to the Government for inspection to determine conformance to the requirements of this specification.

3.2.1 Workmanship and reliability. The workmanship shall be of a quality to assure that the vehicle and components thereof are free from any defects that compromise, limit, or reduce the vehicle or component capabilities herein specified. Vehicles manufactured in accordance with this design shall be subjected to examinations and tests by the Government to proove vehicle capabilities and performance in accordance with the following:

a. Vehicle shall require only organizational maintenance (first and second echelon) during 2,000 miles of normal operation.

b. Vehicle shall require no replacement or major overhaul of any major automotive component during 4,000 miles of normal operation.

c. With the exception of track rubber components, all components of complete track assemblies shall require no more than 10 percent replacement during 2500 miles of operation.

3.2.1.1 Reliability operations. To assure that workmanship and reliability requirements are met during the 4,000 miles of operation, the vehicle combat loaded and with a ROLAND Fire Unit Module or a simulated ROLAND Module shall be capable of operating as follows:

a. 25 percent on paved roads either concrete or asphalt or any combination of the two.
b. 50 percent on gravel and dirt roads with at least 15 percent of this distance over hilly conditions.
c. 15 percent on level cross-country terrain.
d. 10 percent on hilly cross-country terrain.

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3.3 Materials.

3.3.1 Qualified products. Vehicle manufacturer shall be responsible for ascertaining that all assemblies and component parts entering into the construction of this vehicle, which require product qualification in accordance with a specification, are listed on respective qualified products lists (QPL) or have been approved for inclusion on such list. Manufacturer's inspection record shall specifically list such components, name of supplier (s), and number and date of applicable QPL.

3.3.2 Ozone resistant. Rubber products procured under requirements of MIL-R-3065 are to be ozone resistant to the degree specified by the C_1 requirements of MIL-STD-417. Rubber components controlled by detail specifications shall be ozone resistant only when required by the specification referenced on the drawing.

3.4 Construction. Vehicles, components, sub-assemblies, assemblies, shall be fabricated and assembled into a complete vehicle in accordance with drawings listed or referred to in Engineering Parts Lists LP 8750009. All parts requiring identification shall be identified in accordance with MIL-STD-130.

3.4.1 Sealing.

3.4.1.1 Sealer. The sealer used to seal vehicle shall prevent the entrance of water into the hull when vehicle is fording streams or rivers during cross country operation.

3.4.1.2 Hatch, doors, and periscope seals. With hatches and doors closed, the seals shall prevent the entrance of water into the occupied portion of the vehicle.

3.4.2 Special kits. When specified (see 6.2), special kits shall be furnished as follows:

3.4.2.1 Winterization kit. With the winterization kit installed in the vehicle, the engine shall start and perform satisfactory when vehicle is operating in ambient air temperatures minus 25°F to minus 65°F.

3.4.3 Carbon monoxide concentration. The exhaust system shall prevent the accumulation of carbon monoxide in the occupied portions of the vehicle in excess of 0.01 percent under normal conditions of vehicle operation.

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3.4.4 Controls. Controls for the engine, transmission, electrical equipment, permanently mounted fire extinguisher system and other operating mechanisms shall operate without interference.

3.4.5 Throttle linkage. With throttle linkage attached to engine, and pedal fully depressed, the throttle shall be at full rack position. With the throttle at full rack position, the pedal shall be in contact with the stop.

3.4.6 Adjustment. All adjusting mechanisms shall operate without interference and shall maintain adjustment settings under all operating conditions.

3.4.7 Fasteners and locks. All fastening and locking devices shall be positive locking. Means for locking the Driver's Compartment from the outside shall be provided.

3.4.8 Fuel system. During engine operation, the fuel system shall maintain fuel supply to the engine when ascending and descending 60 percent grades in forward and reverse gear, and when vehicle is being operated on 40 percent side slopes, with each side of vehicle up slope.

3.4.8.1 Fuel tanks. Provisions shall be made to assure the internal cleanliness of the fuel tanks prior to initial fueling of vehicle. The fuel tanks shall be capable of receiving fuel at a rate of 50 gallons per minute (gpm), and shall be free from leakage after installation and when tanks are filled to capacity.

3.4.8.2 Fuel lines. Provisions shall be made to assure internal cleanliness of fuel lines and connections prior to final assembly or initial fueling of vehicle. All fuel lines and connections shall be free from leakage.

3.4.9 Electrical system.

3.4.9.1 Generating system voltage (alternator). Prior to vehicle operation, and with the engine operating at 1000 rpm, all vehicle lights on, the generating system shall be set at 28.1 plus or minus 0.1 volts and shall remain within 0.6 volts of the original setting under all conditions of vehicle operation.

3.4.9.2 Lights. The internal and external lights as specified on applicable drawings shall operate throughout all vehicle operating conditions.

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3.4.9.3 Bilge pumps. The electrical system shall supply nominal 24 volts do to the bilge pump motors.

3.4.9.4 Air cleaner blower. The air cleaner motors shall be capable of providing a flow of air through each blower outlet.

3.4.9.5 Personnel heater. The system shall provide sufficient voltage to operate the igniter and heater blower motor.

3.4.9.6 Communication equipment. The communication equipment shall be provided by interconnecting cable from Fire Unit Module.

3.4.10 Welding and weld repairs. All welding and weld repairs shall be in accordance with MIL-W-45205 and MIL-W-45206 as applicable.

3.4.11 Stowed equipment. All on vehicle equipment (OVE) items shall be stowed (combat loaded) on the vehicle in spaces provided without interfering with vehicle operation in any manner.

3.5 Break-in run.

3.5.1 Preparation. Prior to break-in run, except for equipment and component installation and servicing performed by the contractor as a final processing of the vehicle for acceptance, the vehicle shall be completely assembled and all adjustments completed. The vehicle including all components requiring lubrication shall be lubricated as specified in production lubrication chart shown on EPL LP 8750009. The vehicle shall be supplied with fuel conforming to Grade DF 2 of VV-F-800.

3.5.2 Oil pressure and temperature. Prior to the start of the break-in run, the engine shall be operated at idle speed until lubricating oil is at operating pressure and temperature. If at any time during the break-in run the engine has been stopped for at least 30 minutes, the engine shall again be operated, as above, before continuing break-in run. Proper oil pressure and temperature shall be maintained during operating on level ground, 60 percent grades and 40 percent side slope.

3.5.3 Distance operated. Prior to tests, each vehicle shall be given a break-in run on smooth, level, hard-surfaced roads, except a section of the road shall be as shown in Figure 1. The vehicle shall be operated as specified in Table I, and

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over the section of road shown in Figure 1, in "a" and "b" division of the break-in run in both directions for the purpose of observing the function and performance of various parts, assemblies, controls, and suspension system

Table I. Speeds and Distance for Break-in Run

Division of Run	Speed M.P.H.	Distance Miles
a	0 to 10	10
b	11 to 20	25
c	21 to 30	5

3.5.4 Reverse operation. After each division of the break-in run, vehicle shall be stopped, the engine allowed to idle for not less than 2 minutes, and the vehicle driven in reverse for a distance of not less than 50 feet.

3.5.5 Condition after run. After completion of the break-in run, the engine shall idle smoothly. There shall be no defects indicative of damaged components, or maladjustments that may cause faulty vehicle operation.

3.6 Performance. A complete vehicle, combat loaded, with a ROLAND Fire Unit Module or a simulated ROLAND Module, and serviced, shall perform as specified herein. Vehicle; serviced (see 3.5.1) and equipped for existing climatic conditions, shall operate as specified without special equipment. Unless otherwise specified, performance shall be demonstrated on smooth, level, hard-surfaced roads free of loose material, except a section of road shall be as shown in Figure 1.

3.6.1 Environmental. The vehicle shall operate in ambient air temperature of plus 115°F to minus 25°F, and with special equipment installed, at temperature ranges of minus 25°F to minus 65°F. The complete vehicle when in storage shall withstand climatic extremes as specified in MIL-STD-210, without deterioration that may cause failure of any component part of the vehicle.

3.6.2 Power package and drive train. When power package and drive train are installed in accordance with applicable drawings, they shall function throughout all gear and speed ranges without loss of lubricants, and damage that may cause

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failure of vehicle. The power package and drive train and associated controls shall be capable of operation throughout all speed and steering ranges without binding of linkages, loss of lubricants, grabbing, and chattering or slippage when controls are applied to control, stop, and hold vehicle.

3.6.2.1 Cooling system. The engine water temperature, engine lubricant and transmission lubricant temperatures, shall not exceed those shown in Table II when the vehicle is operated under any one or a combination of the following conditions:

a. Ambient air temperature up to 115°F.
b. Under full load.
c. At all speeds down to the track slippage point of the vehicle when operating on smooth, level, hard-surfaced roads, but not less than 3 miles per hour (mph) in first gear or below a transmission speed ratio of 0.56 in other gears.

Table II. Operating Temperatures

Water Temp. from engine	Maximum. engine oil temperature in sump	Max. transmission oil temp. into cooler
23°F	275°F	300°F

3.6.3 Air cleaners. With a maximum air flow of 600 cubic feet per minute, the air cleaner shall be capable of removing not less than 99.8 percent of dust and other foreign material from the engine intake air.

3.6.4 Water removal (bilge pumps). The bilge pumps installed in accordance with applicable drawings shall remove water from the vehicle at a minimum rate of 40 gallons per minute (gpm), with a discharge head of 4 1/2 feet and with pump motor motor operating on 27.5 volts impressed at the motor.

3.6.5 Speeds.

3.6.5.1 Speeds without towed load. Vehicles shall be capable of operating at a sustained speed of 30 mph and shall be capable of maintaining a minimum of 2.5 mph in lowest gear without towed load. The vehicle shall be capable of obtaining a maximum speed of 34 mph. With vehicle traveling between 25 mph and maximum speed, the drift from a straight line shall be not more than 3 feet in 100 feet.

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3.6.5.2 Speeds with towed load. The vehicle shall be capable of towing a track laying free rolling vehicle of equal weight at a sustained speed of 15 mph.

3.6.6 Acceleration. The vehicle on dry, smooth, level, hard surfaced roads free of loose material shall from a standing start be capable of accelerating through a distance of 300 feet in not more than 15 seconds.

3.6.7 Slope operation. Vehicle shall be capable of sustained automotive operation on 40 percent side slopes with each side of vehicle up slope.

3.6.7.1 Engine starting on grades and side slopes. When standing on a 60 percent longitudinal grade, for not less than 2 minutes, with engine operating at idle, the engine shall be stopped for not less than two minutes. The engine shall restart in not more than 1 minute when headed up and headed down grade. A similar engine starting test shall be conducted on 40 percent side slope with each side of vehicle up slope.

3.6.8 Braking.

3.6.8.1 Stopping. Vehicle traveling 20 mph, on dry, smooth, level, hard-surfaced roads free of loose material shall be capable of being stopped by service brake application within a distance of 35 feet from point of brake application. Vehicle drift during stopping shall not exceed three feet.

3.6.8.2 Holding. Vehicle, standing on a 60 percent longitudinal grade, with brakes applied and locked for parking, transmission in neutral, vehicle shall remain stationary when headed up and down grade.

3.6.9 Turning. The vehicle shall be capable of making continuous 360 degree turns to the right and to the left. With the vehicle in geared steer, the rear point or outer edge of vehicle shall be within a 100 foot diameter circle at all times. When clutch brake steer is employed, the vehicle's track pattern shall be within a 35 foot diameter circle at all times.

3.6.10 Fording shallow water.

3.6.10.1 Fording shallow water. Vehicle shall ford hard-botton salt or fresh water crossings of 42 inches in depth, without special equipment. With vehicle operating in water up to 42 inches in depth for 15 minutes, without the aid of bilge pumps, the accumulation of water shall be not more than 1 inch in the bottom of the hull.

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3.6.10.2 Lubricant contamination. After fording for 15 minutes, the water contamination content of the transmission, engine, final drives, and suspension system shall be not more than 2 percent (by volume).

3.6.11 Climbing. Vehicle shall climb a prepared 60 percent longitudinal grade in forward and reverse gear from a standing start on the grade.

3.6.12 Trench crossing. Vehicle shall cross trenches 36 inches in depth and 72 inches in width at low speeds without stalling or damage to vehicle.

3.6.13 Vertical obstacle. Vehicle shall cross vertical obstacles up to 21 inches in height at low speeds without stalling or damage to vehicle.

3.6.14 Radio suppression. Complete vehicle shall be radio interference suppressed in accordance with tactical vehicle requirements of MIL-STD-461.

3.6.15 Fire extinguishers.

3.6.15.1 Fixed. In ambient air temperatures of 72°F, the maximum effort required to discharge the system, either internally or externally, shall be not more than 55 pounds.

3.6.15.2 Portable. The portable fire extinguisher, when installed in accordance with applicable drawings, shall be readily accessible for servicing and use.

3.7 Painting, marking and data plates.

3.7.1 Painting. Unless otherwise specified, the exterior and interior of vehicle, components, assemblies, subassemblies, and parts that require painting shall be prepared and painted in accordance with applicable drawings and Standard MIL-STD-193. Vehicle exterior paint and non-skid paint shall be Forest Green.

3.7.2 Marking. Marking shall be in accordance with MIL-STD-642.

3.7.3 Name, shipping, and identification data plates. The plates shall have a black background, with white letters and letter blanks, except that caution and warning plates shall have red letters and letter blanks. The size of the plate, type and size of lettering and the information to be included shall be in accordance with applicable drawings or as specified by the procuring activity.

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3.8 Workmanship. Workmanship shall be of a quality to assure the vehicle and components free of defects resulting from improper manufacturing or assembly practices and shall conform to requirement herein specified.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assume supplies and services conform to prescribed requirements.

4.1.1 Parts and components. Parts, components and assemblies shall be inspected in accordance with the drawings listed on LP 8750009 and QM 8750009.

4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that practices, method, and procedures of the written inspection plan are being properly applied, and (b) Government production inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have an effect upon the quality of the product, will be immediately called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.3 Qualified products. The contractor's inspection records shall be checked to determine contractors conformance to 3.3.1.

4.2 Preproduction and first production inspection.

4.2.1 Preproduction inspection. A preproduction vehicle (see 3.1) when furnished shall be inspected to ascertain vehicle conformance with the requirements of this specification. Inspection shall be performed at a location approved by

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the Government. Failure of the preproduction model to comply with any of the requirements specified shall be cause for refusal by the Government to conduct a retest until corrective measures satisfactory to the Government have been taken.

4.2.2 First production inspection.

4.2.2.1 Special process inspection. A special process inspection will be conducted by the Government representatives during fabrication of the first production vehicle (see 3.2) to evaluate conformance of materials and workmanship with drawing and specification requirements.

4.2.2.1.1 Review and evaluation. Review and evaluation of records for processing procedure, inspection systems and plan of quality control, will be conducted on the following: material, welding, surface hardening, heat treating, protective finishes, etc.

4.2.2.1.2 Hull inspection. The hull shall be inspected prior to covering the weldments with paint or other pertinent components or assemblies.

4.2.2.2 Complete vehicle inspection. The first vehicle when complete, shall be inspected by the Government subsequent to contractor's inspection at the contractor's plant to determine conformance to contract and specification requirements.

4.2.2.2.1 Corrections. Corrections found necessary as a result of above inspection shall be made by the contractor on the vehicle.

4.2.2.3 Production vehicle test. After completion of the first vehicle inspection, a second vehicle shall be selected from the first month's production or one of the first 10 shall be subjected to examination (see 4.4.2) and tests as specified in 4.5 (a) and (b). Subsequent to examination and tests, the selected vehicle shall be subjected to the 4,000 mile test specified in Table III. These tests shall be performed by the Government, at a site approved by the Government, which will require a maximum of 90 days. Delays caused by vehicle break-down due to poor quality of workmanship or material, or failure of the contractor to comply with specifications or drawing requirements shall not be the basis for adjustment of the contract performance dates or delivery schedule or contract price. Under separate contractual authority, the contractor will expeditiously furnish repair parts, as required to adequately support the above tests. Delays caused by vehicle break-down and failure of the contractor to adequately support the vehicle with parts during the tests shall not be the responsibility of the Government.

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Table III. 4000 Mile Test, Combat Loaded, with a ROLAND Fire Unit Module or a Simulated ROLAND Module. (Government Proving Grounds)

Course	Mileage and Speeds
Hard surface roads	1000 mi. at varying speeds up to maximum.
Gravel & dirt roads	2000 mi. at varying speeds up to maximum with 300 miles of this distance over hilly conditions.
Level cross-country	600 mi. at varying speeds up to maximum.
Hilly cross-country	400 mi. at varying speeds up to maximum.

4.2.3 Test failure. Failure of the vehicle to comply with any of the requirements specified or any deficiency of a workmanship or materials nature found on the vehicle during or as a result of the 4000 mile test, will be cause for rejection of the vehicle. Further, the Government will refuse to continue acceptance of production vehicles until evidence has been provided by the manufacturer that corrective action has been taken to eliminate the deficiency. Any deficiency found during or as a result of 4000 mile test shall be prima facie evidence that all vehicles already prior to completion of the 4000 mile test are similarly deficient, unless evidence satisfactory to the contracting officer is furnished by the contractor that they are not similarly deficient. Such deficiencies on all vehicles shall be corrected by the contractor at no cost to the Government.

4.3 Break-in run, Track. To determine conformance to 3.5.1, 3.5.2, and 3.5.3 each vehicle shall be lubricated and serviced and warmed up for not less than twenty minutes prior to the break-in run. Records shall be kept of the vehicle instrument readings. The run shall be on smooth, level, hard-surface roads, and for the distance specified in Table I. The vehicle shall be operated in reverse after each division as specified in 3.5.4. The undulating portion of the test shall be accomplished twice in each direction at speeds between 5 and 10 mph during division "a" and between 11 and 14.5 mph during division "b". The results of the break-in runs on the first twenty vehicles, that meet all requirements, will serve as a standard for the evaluation of the balance of production.

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4.3.1 Condition after run. After the break-in run, the vehicle shall be inspected to determine conformance to 3.5.5.

4.4 Inspection provisions.

4.4.1 Lot size. An inspection lot shall consist of all vehicles of one type, from one day's production, submitted at one time for acceptance inspection.

4.4.1.1 Sampling for inspection. For the purpose of visual, dimensional, and primary functional inspection, a representative sample shall be selected from each inspection lot in accordance with inspection level II of MIL-STD-105. Before sampling may be accomplished by the contractor, the contractor shall 100 percent inspect the first 20 vehicles to establish a process average (see 6.5) to allow normal sampling in accordance with MIL-STD-105.

4.4.2 Examinations. Visual, dimensional, and primary functional examination shall consist of examination of the vehicle for conformance to the applicable drawings and this specification. Examination shall be performed against the classification of defects and with the acceptable quality levels shown in Table IV. The following constitutes a part of the classification of defects (see 6.4):

a. Any drip constitutes a major defect when the vehicle has been standing idle and the components are at ambient temperature.

b. Any leak that occurs at a static fit, metal to metal or gasket combination shall constitute a major defect.

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Table IV Classification of Defects

Characteristics	Major: AQL 25 Defects/100 Units	Minor: AQL 150 Defects/100 Units	Method of Inspection
101. Engine	Malfunction, leaks, (see 6.4) (b), mounting improper, damage	201. Improper component assembly or installation	Visual and functional
102. Engine governors	Improper setting, malfunction	202. Missing or defective seals	Functional
103. Power train	Malfunction, mounting improper damage, and leakage (see 6.4) (b)	203. Improper component assembly or installation	Visual and functional
104. Fuel system components	Malfunction, damage, leaks (see 6.4) (a)	Improper assembly, installation or clearance	Visual and functional
105. Fuel and oil lines valves and hose	Defective, clearance improper malfunction, leaks (see 6.4) (a)	205. Improper assembly or installation	Visual and functional
106. Cooling system components (engine and transmission)	Malfunction, damage, leaks (see 6.4 (b)	206. Improper assembly or installation	Visual and functional
107. Exhaust system components	Damage, leaks	207. Improper assembly, installation or clearance	Visual
108. Electrical system *components, lights (Include wiring and instrumentation)	Malfunction, damage	208. Improper assembly, installation, circuit identification, clearance or protection, illumination of reeicle and cant connector level vial	Visual functional
109. Suspension system components	Damage, tire or track pad separation, improper clearance or alignment, welding defects, leaks, (see 6.4) (b), malfunction	Improper assembly, installation bolt or not torque and track tension.	Visual
110. Hull	Structure and welding defect		Visual

* NOTE: Filament failures are not be to classified as a major or minor defect.

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Table IV Classification of Defects (Cont'd)

Characteristics	Major: AQL 25 Defects/100 Units	Minor: AQL 150 Defects/100 Units	Method of Inspection
111. Control and adjusting mechanisms	Malfunction, improper clearance	210. Improper assembly or installation, welding defects	Visual and functional
112. Doors, fenders, grilles, panels ducts, stowage brackets, boxes, racks and stops	Malfunction: torsion spring and hold-open devices	211. Improper fits, adjustment, assembly or installation	Visual and functional
113. Cushion, seat, rest and hatch seals	Seat control malfunction, damage, leakage (see 6.4) (d)	212. Improper assembly or installation, adhesion damage	Visual and functional
114. Misc items accessories	Malfunction, damage, missing	213. Improper assembly or installation	Visual and functional
115. Fire extinguisher system components	Lockwire missing	214. Data card incomplete, improper assembly or installation	Visual and functional
116. Speed	Nonconformance		Functional
117. Acceleration	Nonconformance		Functional
118. Turning ability and steering	Nonconformance, wander, side-pull, unusual noise		Functional
119. Braking ability (parking and service brakes)	Nonconformance: grade holding or stopping distance, equalization improper, unusual noise		Functional
120. Fording (shallow)	Water leakage excessive		Visual
121. Paint protective and anti-skid coatings		215. Application, coverage and materials improper	Visual
122. Lube fittings and lubrication		216. Fittings missing, defective or improperly installed, lubrication improper	Visual

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4.4.2.1 Unclassified defects. All defects having no bearing on function, safety, interchangeability or life, but are considered departures from good workmanship will be noted in writing. Workmanship deficiencies falling within this category and recurring in 5 consecutive lots or ten lots or more within a thirty-day period will be added to the minor classification with no increase to AQL's.

4.4.2.2 Recurring major deficiencies. A major deficiency (see 6.6) is recurring when the same defect occurs more than once in the same sample or when the defect occurs in two successive samples. A major defect may be considered recurring when the historical inspection records ("P" chart or approved equivalent) reflect such a condition. Recurring major deficiencies shall be cause for the entire lot or lots to be inspected for the reeurring deficiencies. The deficiencies shall be corrected when found.

4.4.2.3 Recurring minor deficiencies. A minor deficiency (see 6.6) is recurring if it occurs more than twice in the same same sample or when the defect occurs in four successive samples. Recurring minor deficiencies shall be cause for the entire lot or lots to be inspected for the recurring deficiency. Correction of all deficiencies shall be accomplished prior to acceptance.

4.5 Classification of tests.

a. Acceptance tests, 4.5.1.1 through 4.5.1.20
b. Control tests, 4.5.2 through 4.5.2.7
c. Inspection comparison test (ICT), 4.5.3.2 through 4.5.3.13.

4.5.1 Tests.

4.5.1.1 Acceptance tests. To determine conformance to 3.6, after the break-in run (see 3.5) each vehicle shall be operated for a distance of eight miles or more and shall be inspected as specified in 4.4.2. These acceptance tests shall include all acceptance tests specified at the place of manufacture in Table V. Vehicles shall be completely assembled and serviced, with fuel, lubricants, and fluid as specified, but need not be combat loaded, but shall be loaded with a ROLAND Fire Unit Module or a simulated ROLAND Module. After the tests, the vehicle shall be examined for evidence of fuel and lubricant leakage, or other deficiencies.

4.5.1.1.1 Test failure. If a vehicle fails to pass any acceptance test specified herein, the Government inspector shall stop acceptance of subsequent lots until evidence has been provided by the contractor that the corrective action has been taken.

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Table V. Location for Tests

Title	Paragraph Numbers	Place of Manufacture	Government Proving Ground
Acceptance tests	4.5.1.1	X	X
Sealing and seals check	4.5.1.2	X	X
Controls and linkage check	4.5.1.3	X	X
Adjustment mechanism check	4.5.1.4	X	X
Fastener and lock check	4.5.1.5	X	X
Fuel system leakage check	4.5.1.6	X	X
Generating system voltage check (alternator)	4.5.1.7	X	X
Light check	4.5.1.8	X	X
Bilge pumps electrical check	4.5.1.9	X	X
Air cleaner blower electrical check	4.5.1.10	X	X
Personnel heater electrical check	4.5.1.11	X	X
Welding and weld repair check	4.5.1.12	X	X
Power package and train test	4.5.1.13	X	X
Speed test	4.5.1.14	X	X
Acceleration test	4.5.1.15	X	X
Engine starting on grades and side slope tests	4.5.1.16	X	X
Stopping tests	4.5.1.17	X	X
Holding test	4.5.1.18	X	X
Turning test	4.5.1.19	X	X
Shallow water fording test	4.5.1.20	X	X
Control tests	4.5.2	X	X
Frequency	4.5.2.1	X	X
Contamination test	4.5.2.3	X	X
Fuel system slope test	4.5.2.4	X	X
Fuel tank and line test	4.5.2.5	X	X
Stowed equipment test	4.5.2.6	X	X
Fire extinguisher test	4.5.2.7	X	X
Inspection comparison tests	4.5.3		X
Winterization equipment test	4.5.3.2		X
Carbon monoxide concentration test	4.5.3.3		X
Temperature operational tests	4.5.3.4		X
Cooling system tests	4.5.3.5		X
Air cleaner operational check	4.5.3.6		X
Water removal (bilge pumps) test	4.5.3.7		X
Speeds with towed load test	4.5.3.8		X
Slope operational test	4.5.3.9		X
Climbing test	4.5.3.10		X
Trench crossing test	4.5.3.11		X
Vertical obstacle test	4.5.3.12		X
Radio Supression test	4.5.3.13		X

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4.5.1.2 Sealing and seals check. To determine conformance to 3.4.1 through 3.4.1.2, the sealing, hatch doors, periscope, compartment doors shall be checked for leakage of water into the hull.

4.5.1.3 Controls and linkage checks. To determine conformance to 3.4.4 and 3.4.5, all controls, and throttle linkage shall be operated and checked for functional requirements.

4.5.1.4 Adjustment mechanism check. To determine conformance to 3.4.6, all adjustment mechanisms shall be checked for proper adjustment and functional requirements.

4.5.1.5 Fastener and lock check. To determine conformance to 3.4.7, all fastener and locking devices shall be operated and checked for positive fastening and locking.

4.5.1.6 Fuel system leakage check. To determine conformance to 3.4.8.1 and 3.4.8.2, prior to engine installation, the fuel system shall be pressurized with dry air at a pressure of 3 to 4 pounds per square inch (psi), and check fittings, lines, and fuel tank seals for leaks by applying soapy water or equivalent.

4.5.1.7 Generating system voltage check (alternator). To determine conformance to 3.4.9.1 prior to performance tests, the generating system voltage shall be checked with master volt meter, and the voltage shall be within plus or minus 0.6 volts of the original setting.

4.5.1.8 Light check. To determine conformance to 3.4.9.2, all lights shall be operated during and after break-in run throughout all stages of operation, and checked for functional requirements.

4.5.1.9 Bilge pumps electrical check. To determine conformance to 3.4.9.3, the bilge pumps electrical controls shall be turned on and the pumps checked for operation.

4.5.1.10 Air cleaner blower electrical check. To determine conformance to 3.4.9.4, the air cleaner electrical controls shall be turned on and the blower checked for operation and air flow.

4.5.1.11 Personnel heater electrical check. To determine conformance to 3.4.9.4, the personnel heater electrical controls shall be turned on and the heater checked for operation.

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4.5.1.12 Welding and weld repair check. To determine conformance to 3.4.10, all welding and weld repairs shall be examined for possible defects and repairs accomplished in accordance with approved procedure.

4.5.1.13 Power package and drive train test. To determine conformance to 3.6.2, the vehicle shall be operated throughout all gear and speed ranges, the power plant, power train, final drives, tracks, suspension system, and applicable controls shall be checked for functional requirements.

4.5.1.14 Speed tests without towed load. To determine conformance to 3.6.5.1, the vehicle shall be operated at specific speeds and checked for functional requirements.

4.5.1.15 Acceleration test. To determine conformance to 3.6.6, the vehicle shall be accelerated from a standing start to the required speeds and the time checked.

4.5.1.16 Engine starting on grades and side slope tests. To determine conformance to 3.5.2 and 3.6.7.1, the engine shall start, oil pressure and temperature shall be maintained when operating on specified grades and slopes. The 60 percent grade test may be conducted in conjunction with brake holding test.

4.5.1.17 Stopping test. To determine conformance to 3.6.8.1, the vehicle shall be operated on a dry level hard-surfaced road free of loose material, and checked for functional requirements.

4.5.1.18 Holding test. To determine conformance to 3.6.8.2, the vehicle shall be driven up a specified grade, service brakes applied and vehicle observed for movement, then with parking brakes locked the vehicle shall be observed for movement.

4.5.1.19 Turning test. To determine conformance to 3.6.9, vehicle shall be operated and turned to right and left as specified and checked for radius of turns.

4.5.1.20 Shallow water fording test. To determine conformance to 3.6.10.1, the vehicle shall be driven into water of specified depth, without special equipment. Water accumulation shall not exceed specified depth in hull bottom after 15 minutes of fording operation. The engie shall idle at 1000 rpm for 15 minutes, be stopped, and shall restart in not more than 3 minutes with all accessories functioning satisfactorily during and after fording.

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4.5.2 Control test. Control tests for maintaining control of those requirements whose inspection would be of too long duration to be included under acceptance tests shall be conducted by the manufacturer as specified herein.

4.5.2.1 Frequency. The first vehicle off the production line, or one of the first ten vehicles produced, and thereafter one per month, shall be checked in accordance with the control tests.

4.5.2.2 Test failure. If a vehicle fails to pass any control test specified herein, the Government inspector shall stop acceptance of subsequent lots until evidence has been provided by the contractor that corrective action has been taken.

4.5.2.3 Contamination test. To determine conformance to 3.6.10.2, the vehicle after the water fording test shall be examined for contamination of lubricants of the transmission, engine, final drives, and road wheels 2, 3, 4, 5, and one of their corresponding arms, on each side of the vehicle. Contamination shall not exceed that specified.

4.5.2.4 Fuel system slope test. To determine conformance to 3.4.8, with vehicle headed up and down a 60 percent longitudinal slope, and each side up on a 40 percent slope, vehicle shall be checked for functional requirements.

4.5.2.5 Fuel tank and lines test. To determine conformance to 3.4.8.1 and 3.4.8.2, the fuel tanks shall be filled to capacity at a rate of not less than 50 gallons per minute. The tanks and lines shall be checked for leaks before and after a two mile road test on smooth, level, hard-surfaced roads.

4.5.2.6 Stowed equipment test. To determine conformance to 3.4.11, all manufacturer and depot installed "On Vehicle Equipment" (OVE) shall be stowed (combat loaded) on the selected vehicle. The equipment shall be removed after the test. The OVE shall be made available to the contractor, and the contractor shall be permitted to unpackage and repackage for domestic shipment only the equipment used for test.

4.5.2.7 Fire extinguisher test. To determine conformance to 3.6.15.1, the fixed fire extinguisher system shall be functionally tested, both inside and out, subsequent tests of one in each fifty testing alternately.

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4.5.3 Inspection comparison tests. The Government may select vehicles any time during the contract production period and subject vehicle to tests listed in Table V and Table VI to reveal deficiencies of manufacturer or design that may reduce the effective operation of these items in the field and to compare existing quality with previous standards. These tests shall be conducted at Government laboratories or proving grounds designated by the contracting officer. Selection of vehicles shall be on a spot check basis. Comparison test vehicles shall be combat loaded with all on-vehicle equipment stowed in their applicable spaced to conform to 3.4.11. Vehicle selected shall not include any vehicles previously tested for conformance to 4.5.2 (control tests).

4.5.3.1 Test failure. Failure of any vehicle tested as specified in 4.5.3 to comply with any of the requirements specified in the contract, or any major deficiency of a workmanship or materials nature occurring during, or as a result of the test, may be cause for refusal to continue acceptance of vehicles by the Government until evidence has been provided by the manufacturer that corrective action has been taken to eliminate the conditions.

Table VI: 2000 Mile Test Combat Loaded with a ROLAND Fire Unit Module or a Simulated ROLAND Module (Government Proving Ground)

Course	Mileage & Speeds
Hard surface roads	200 mi. at varying speeds up to maximum
Gravel & dirt roads	500 mi. at varying speeds up to maximum w/75 mi. of distance on hilly conditions
Level cross country	650 mi. at varying speeds up to maximum
Hilly cross country	650 mi. at varying speeds up to maximum

4.5.3.2 Winterization equipment test. Winterization equipment shall be installed in accordance with applicable drawings and specifications. The vehicle, properly serviced and equipped, shall be stabilized and then subjected to respective ambient temperature of minus 25°F to minus 65°F. The engine shall satisfactorily start and operate and the personnel heaters shall function as specified in 3.4.2.1.

4.5.3.3 Carbon monoxide concentration test. To determine conformance to 3.4.3, the driver's compartment shall be checked for carbon monoxide concentration.

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4.5.3.4 Temperature operational tests. To determine conformance to 3.6.1, the vehicle, properly serviced and equipped, shall be subjected to the applicable tests in high and low temperatures.

4.5.3.5 Cooling system tests. To determine conformance to 3.6.2.1, the vehicle shall be operated as specified and cooling systems checked for functional requirements.

4.5.3.6 Air cleaner operational check. To determine conformance to 3.6.3 during and after operation, the air cleaners shall be checked for functional requirements.

4.5.3.7 Water removal (bilge pumps) check. To determine conformance to 3.6.4, the vehicle shall have sufficient water in hull to operate bilge pumps and check the rate of water removal with the specified discharge head.

4.5.3.8 Speeds with towed load test. To determine conformance to 3.6.5.2, the vehicle shall be operated towing specified vehicle, at indicated speeds for a distance of not less than that required to meet performance requirements.

4.5.3.9 Slope operational test. To determine conformance to 3.6.7, the vehicle shall be operated as specified, on applicable slopes and checked for performance requirements.

4.5.3.10 Climbing operational test. To determine conformance to 3.6.11, each vehicle selected shall be driven in forward and reverse gear up approved 60 percent grade.

4.5.3.11 Trench crossing tests. To determine conformance to 3.6.12, the vehicle selected shall be driven without stalling over trenches as specified and then examined for damage.

4.5.3.12 Vertical obstacle tests. To determine conformance to 3.6.13, the vehicle selected shall be driven without stalling over obstacles as specified and then examined for damage as specified.

4.5.3.13 Radio suppression test. To determine conformance to 3.6.14, the vehicle shall be subjected to a radio interference suppression test.

5. Packaging.

5.1 Preparation for delivery and storage shall be in accordance with MIL-H-46709.

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6. NOTES

6.1 Intended use. The vehicle is intended for use by the Armed Forces as a Carrier Vehicle for the U.S. ROLAND Fire Unit Module.

6.2 Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Preproduction model if required (see 3.1)
c. Special kits if required (see 3.4.2).

6.3 Safety precautions. Caution should be exercised in handling carbon dioxide fire-extinguisher cylinders. They should not be dropped, permitted to strike each other, or be handled roughly. Extreme care should be exercised during reinstallation to avoid tripping fire-extinguisher control system since physical injury is highly probable.

6.4 The following definitions shall be used for the classification of defects for leaks:

a. Weep: - Any evidence of fluid beyond the seal.
b. Seep: - Any evidence of fluid beyond the seal that does not result in formation of a droplet.
c. Droplets: - Any evidence of fluid beyond the seal that results in the formation of a droplet.
d. Drip: - Any evidence of fluid beyone the seal where droplet forms and falls.

6.5 Process average. Sampling may be initiated if the process average value for the first twenty vehicles inspected in less than the AQL specified in the classification of defects for major and minor defects. Process average shall be computed as follows:

$$\text{Process Average} = \frac{\text{Number of Defects}}{\text{Number of Vehicles Inspected}} \times 100$$

If the computed process average exceeds the specified AQL, 100% inspection shall be performed and continued until such time that the process average for twenty consecutive vehicles is less than the specified AQL.

6.6 Definitions of recurring major and minor defects. The Government Inspector will verify that a thorough inspection of each vehicle is performed by the contractor not only for the

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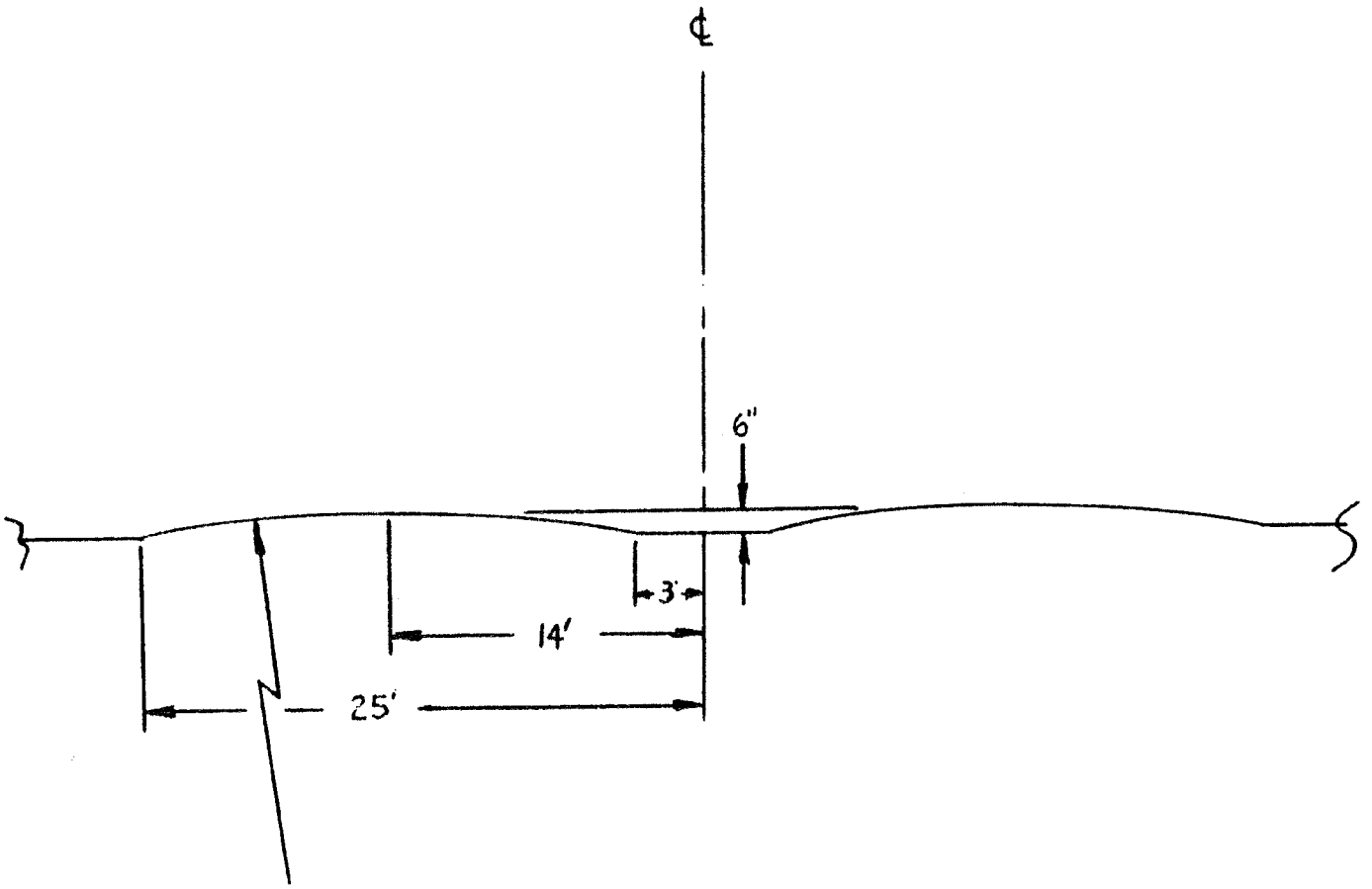


FIGURE 1 - PROFILE OF TEST TRACK
(ARTIFICIAL BUMP)

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listed characteristics, but also for any other departures from good workmanship. The Government Inspector will assure that all deficiencies encountered during the inspection are enumerated on the Deficiency Sheets for the vehicle. The defects noted on the Deficiency Sheet shall contain sufficient description to enable the Government Inspector and the contractor's representative to classify the deficiency in accordance with the classification of defects of the vehicle specification and deficiencies contained in MIL-STD-105. Corrective action shall be taken for recurring deficiencies.

6.7 Inspection equipment calibration and inspection. In inspecting the supplier's inspection equipment, the Government inspector will determine that the supplier has available, and utilizes correctly, gaging, measuring, and test equipment of the required accuracy and precision. Except as otherwise specified, the supplier will have available a set of master gages, standards, and appropriate instruments for regularly scheduled calibration of his inspection equipment. Records of such regularly scheduled calibration will be maintained by the supplier and made available for review by the Government. The calibration of gages, standards, and instruments will be periodically checked by authorized Government personnel.

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Army-AR

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